

A
Project Report
on

**“DELINEATION OF WATERSHED AND ESTIMATION OF
DISCHARGE OF RIVER TAPI SUB-BASIN BY USING GIS TOOLS”**

SUBMITTED TO



SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

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UNDER THE GUIDANCE OF

Prof. K. A. Salunke



DEPARTMENT OF CIVIL ENGINEERING

SANDIP FOUNDATION'S

SANDIP INSTITUTE OF ENGINEERING AND MANAGEMENT,

NASHIK.

AY- (2022-23)

SANDIP FOUNDATION'S
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CERTIFICATE

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**This work is submitted in the partial fulfillment of the prescribed syllabus of
Savitribai Phule Pune University, Pune
For the academic year 2022-23**

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**“DELINEATION OF WATERSHED AND ESTIMATION OF DISCHARGE
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In partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering** in **Civil Engineering** of **Savitribai Phule Pune University, Pune** is hereby approved for the award of the degree.

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Acknowledgement

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CHAPTER 1

INTRODUCTION

1.1 General

Water is one of the most requisite resources for all forms of life. In India due to the rapid growth of industrialization, urbanization and populism, sources of water go on decreasing. So, it is essential to conserve restore and these water resources in an effective way.

For this, there is a need of GIS software that provides management, planning and analysis of water resources and with the help of maps, spatial data and various hydrological parameters, GIS software performs hydrological modelling which will be useful for the management program. QGIS is one of the most powerful GIS applications. By supporting both raster and vector database functionalities, this software helps to Analyze, edit, compose capture, store various file formats which contain spatial information. There are various applications of the QGIS tool for e.g., water and wastewater network management, electricity and natural gas network management, transport and road mapping, urban planning, flood management, health / medical resource management, etc.

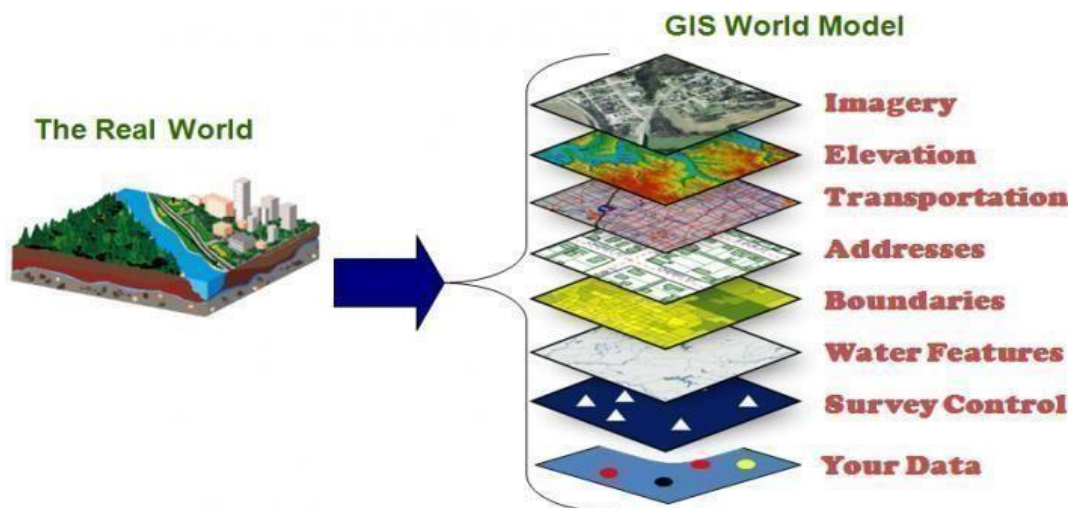


Fig 1.1: GIS World Model

Among all these applications, GIS has a vital role in watershed delineation and watershed management. Watershed delineation is the process of identifying the drainage area of any point on a stream or river network. Topography (topographical maps) is usually the main input in determining a river watershed therefore its delineation requires the use of topographic maps which are sometimes not easily available and outdated in most places, including India. They are mostly of different scales when more than one sheet of topographic map is needed to cover the area of interest. By using topographical maps watershed can be delineated manually but this manual work is a very tedious and difficult task especially in flat terrains. Digital Elevation Models (DEM) and Geographic Information System (GIS) software are tools that can be used for modelling of stream networks, delineation of watershed to obtain parameters that are important for management of water volume and quality, soil conservation, flood control, wildlife habitat and other hydrological analyses. The quality and type of DEM along with its computer algorithms define the accuracy of result.

In this study, our aim is to use the GIS tool for the watershed delineation of the Tapi river basin. GIS software provides watershed delineation which can give factors for calculation of rainfall-runoff model and delineate a boundary for a catchment area. In our study by performing various algorithms in QGIS tools, this tool generates a thematic map and DEM model along with the output parameters. After performing a comparative analysis between QGIS and competent authority data, the accuracy of QGIS software will be studied and concluded.

1.2 Importance of the Tapi river:

Tapi has several ports where international trade occurs. Tapi is one of the rivers that flows through three leading states in India. It has high-quality and richly fertile soil that is best for agricultural activities. The river is also heavily used for irrigation reasons. Tapi is home to the various natural habitats of several wild and exotic animals. Tapi includes up to 35+ irrigation projects which also include the world's second- largest dam – the Sagar dam.

1.3 Scope of GIS tools to delineate the watershed.

One of the fastest-growing, most exciting and powerful systems in today's world is the geographic information system. GIS is one of the best customized and relative tools in the world as we can perform various tasks from basic mapping to the planning and administration of transportation, utility infrastructure, urban development, disaster management, wastewater management and storm water management.

From all these applications GIS is a very powerful geomatics decision-making tool for the planning, monitoring and development of watershed areas. The study of watershed delineation using the QGIS tool will be the best reference for the following parameters: -

- Alternate land use practice
- Selecting sites for water harvesting structures
- Adoption of water conservation measures
- Making policy by local government authorities related to energy, water resources, irrigation and various other infrastructures.
- Government to build a new drainage network, rainwater harvesting structures, roadwork network and disaster management network.
- A detailed estimation of cost for rejuvenation of streams of various orders

1.4 Problem statement:

- Manually watershed management is more time-consuming so to reduce the time we need to plan watershed management with the help of GIS software. GIS software is easy to use and give more satisfactory result in a short time as compared to manual planning.
- As we are going to perform watershed delineation of Tapi river basin by using QGIS tools, we will get the best results with a minimum error by performing data analysis in QGIS.

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1.5 Aim of the project:

- To delineate the watershed of a Tapi river basin.
- To estimate the total discharge of the Tapi river basin.
- Comparative analysis of watershed delineation between QGIS and competent authority data.

1.6 Study Area:

Physical Characteristics

- State – Madhya Pradesh, Maharashtra, Gujrat
- Catchment area – 65,145 km²
- Length – 700 km
- Basin extent longitudinal - 72° 38’
to 78° 17’ East and latitude 20° 05’
to 22° 03’ North
- Three main sub basin - 1) Upper Tapi Basin up to Hathnur
2) Middle Tapi basin
3) Lower Tapi basin
- Tributaries - 14
- Major tributaries
of Tapi river - Purna river, Girna river, Gomai , Panzara,
Pedhi, Arna

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Table 1.1: Study area of project

Sl. No	Name of State	Drainage area (sqkm)	Percentage of total
1	Madhya Pradesh	9,804	15.0
2	Maharashtra	51,504	79.1
3	Gujarat	3,837	5.9
Total		65,145	100.0

1.7 Indian scenario of GIS software:

In worldwide QGIS software is considered one of the most popular and best software for analyzing geospatial data, United States, United Kingdom and Canada are the top three countries that use QGIS. Around 164 companies in India are using QGIS. In the list of distribution of companies using QGIS by country, India has 6th rank. In India, the use of GIS software by professionals is increasing day by day. So many platforms in India include the uses of GIS as crime mapping, delineation of the watershed, water and wastewater network management, hydrological modeling and simulation, land management system and many more.

1.8 Theoretical Background

This chapter includes the concept of hydrological modeling, different methods of water delineation, and the correlation between hydrological modeling and water delineation. This chapter also describes the basic principles of GIS tools along with their features.

1.9 Technical Parameters

1.9.1 Watershed Delineation:

A watershed is defined as the land region or a catchment area where all the water in the form of rivers, streams or lakes drains to a common outlet together. Smaller watersheds together form a large watershed which is called a river drainage basin.

Watershed delineation is the process of identifying the drainage area or a watershed boundary on a topographical map. Watershed delineation is the best initial step for the future development and the effective management of the particular land.

1.9.2 Stream order:

Stream order is a method of classifying a stream network by assigning numeric order to the links. The increase in the stream order indicates an increase in discharge and a decrease in gradient. The size of stream order ranges from the first order (smallest) stream to the 12th order stream (largest). As this stream order gives the idea of the size and strength of specific waterways within stream networks, it is important to geographers, hydrologists and other scientists.

1.9.3 Raster and Vector data:

Raster and vector are the two spatial data formats in GIS to store geospatial data.

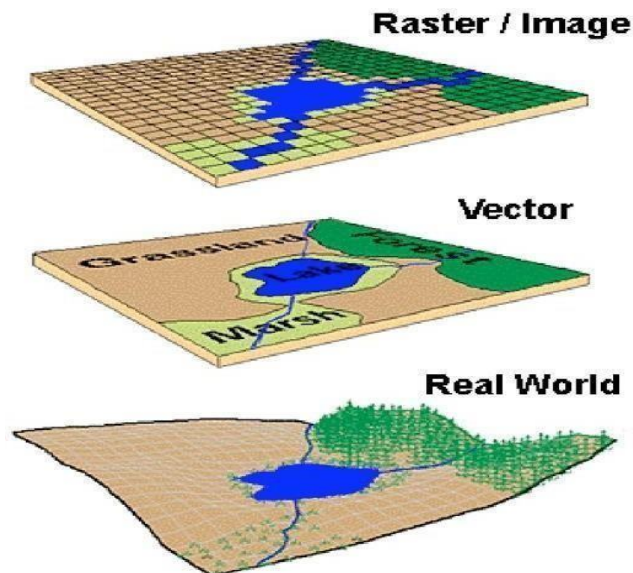


Fig 1.2: Raster and vector image

Raster data / Raster data structure / Raster model: Raster data structure is a geographical entity that is stored in a matrix of rectangular cells or a matrix of square areas which is often called pixels, cells and grids. A code is given to each cell which informs users which entity is present in which cells and these cells describe interiors rather than boundaries. Raster includes digital aerial photographs, satellite images, DEM, digital pictures, survey sheets or scanned maps. Raster data is best when it comes to the more aligned representation or some other physical phenomena or if there is a mathematical context.

Image formats of raster data: PNG, JPG, BMP, GIF, EPS, SVG, PDF

1.9.4 Vector data / Vector data structure / Vector model:

A vector-based GIS is defined by the vectorial representation of its geographic data. The three main geometric shapes used in the vector data model to represent real-world features are:

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Points
Lines
Polygon

It is comprised of points, lines, arcs, circles, polygons, etc. Vector data is extremely useful for storing and representing data that has discrete boundaries.

Such as borders, building footprints, streets and other transport links and location points. Vector data is best when it comes to cartographic representation.

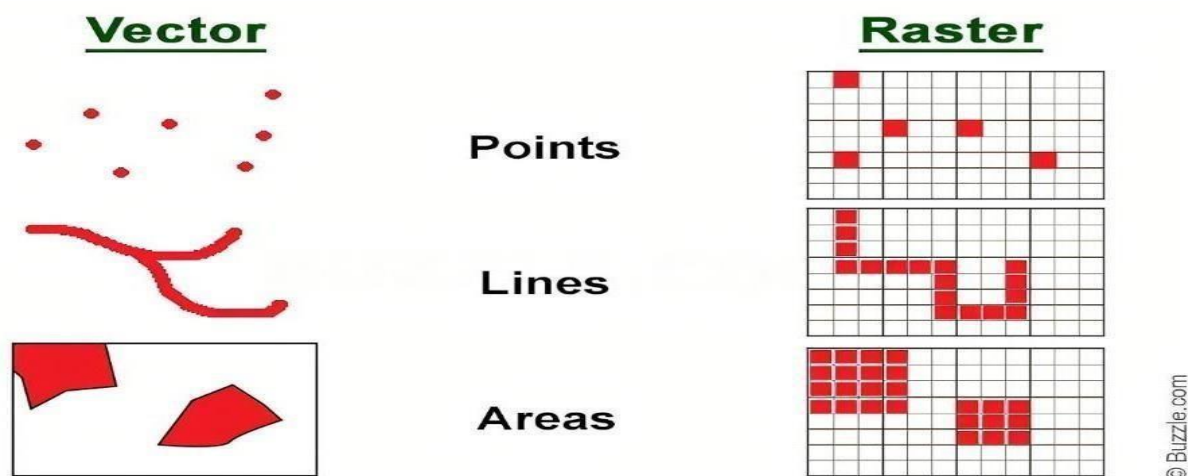


Fig 1.3: Rater vs Vector representation

1.9.5 Hydrological Modelling:

Hydrological modeling simply included characterization of real hydrological features and mathematical representation of flow characteristics. Flow characteristics include four basic categories namely simulation basis, spatial representation, temporal representation, and method of simulation.

1.10 Perspectives of hydrological modelling in GIS:

A hydrological model can generate streamflow estimates and their constituents over the land surface. A model can also be used to generate future yields or scenario modelling. GIS provides representations of these spatial features of the Earth, while hydrologic modelling is concerned with the flow of water and its constituents over the land surface and in the subsurface environment.

1.11 The necessity of water delineation and hydrological modelling:

The determinations and various output parameters derived after a watershed is delineated are essential for the future planning process. The manual procedure of watershed delineation is time-consuming and tedious work so there is a need for tools like QGIS, ArcGIS, GrassGIS or any other GIS tool which will offer effective work management that consumes less time and gives the best results.

The GIS system for watershed delineation and hydrological modeling offers the best planning, execution and monitoring of the project by helping us to build a better understanding of a water quality problem.

1.12 Basic characteristics and features of GIS tools:

- QGIS offers vector analysis, raster analysis, sampling, geoprocessing, geometry and database, management tools.
- We can compose, analyze, view, and explore data.
- The spatial location of features is represented by points, lines or polygons and defined by coordinates in a specific reference system and the geometry of the features refers to the three-dimensional representation in space.
- Ability to create and edit shape files and GRASS vector layers, partial database tables from shape files with DB Manager plugin.
- We can publish our data on the Internet using a webserver with UMN Map Server Or GeoServer installed.

1.13 The stepwise procedure of watershed delineation using QGIS:

1. Preprocessing and downloading input files and DEM files.
2. Obtain depression less DEM.
3. Stream definition and stream segmentation using Strahler stream order.
4. Determine flow direction and flow accumulation.
5. Catchment Grid delineation.
6. Catchment polygon processing
7. Drainage line processing
8. Adjoint catchment processing
9. Drainage point processing
10. Watershed processing

1.14 Input Data:

1. Collection of toposheet, shapefiles
2. Collection of satellite images, KML/KMZ files
3. SRTM data
4. Vector data collection
5. DEM files from google earth.

1.15 Output Data:

1. Various thematic maps such as slope map, hill shed map, aspect map, contour map, drainage density map, pre-monsoon depth of groundwater map, rainfall distribution map.
2. Stream and catchment delineation.
3. Percentage spatial distribution of various parameters of the basin e.g., water provision, soil retention, habitat quality, soil erosion, ecological redline areas.
4. Spatial characteristics of supply, demand and supply-demand balance of water provision service.
5. Stimulation of the runoff data.

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6. The annual runoff data.
7. Discharge of river.
8. Number of water resources, rivers, lakes

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Survey

Creating and analyzing the digital terrain model of the SLIVOVO area using QGIS software—Taylor and Francis- annals of GIS- Volume 43, 2017.

Creating and analyzing the digital terrain model of the Slivovo area using QGIS software. As we know that water is a vital source for our daily life so expanding the Digital Terrain Model (DTM) through QGIS software is the goal of the respective research paper. So as to achieve progress of the research paper we need to test some methods and techniques which are well known in day-to-day life also they are carried out by QGIS software TIN and GRID are the methods mentioned above. The QGIS software has many advantages like speed, performance, proper documentation, compatibility and the DTM in individuals.

A QGIS plugin to tailor SWAT watershed delineations to the lake and reservoir Waterbodies-Environmental modelling and software, Department of bioscience, Aarhus University, Denmark, 2018. During the last century, the quality of water has been degraded because of multiple actions like climate, the increase in water abstractions and the high nutrient level caused by agricultural expansion. In SWAT user interface delineation of the watershed area draining into a lake requires users to draw the watershed outlet in a specific point of the network. And this problem has not yet been solved on SWAT. The areas in between Rivers have been ignored.

This paper describes the development and design of SWAT2LAKE which have not yet been considered before.

GIS-based soil maps as tools to evaluate land capability and suitability in a coastal reclaimed area- International soil and water conservation research, Science direct journal, 2 December 2021. The study to build land capability and suitability maps are useful for soil resource conservation using a multi-thematic approach by GIS in coastal areas of Italy. Hence, the users could help to define best agricultural practices to apply in order to preserve soil functions. Thus, this model is applied to the study where agricultural productivity and environmental impact are in dispute and these maps should never be considered as permanent.

GIS tool for cost-effective delineation of the flood-prone area- Computers and Urban Systems, Elsevier journal, 2018.

Floods are the costliest and the most frequently occurring natural hazards which have killed over 1,75,000 people and affected more than 2.2 billion throughout the world. Despite such advanced computational techniques, flood hazard maps are still lacking in many countries due to the high and expensive cost it takes. Given these assumptions, in this paper, they have developed a practical and cost-effective method to delineate the flood-prone areas in poor locations for large scale based on easily available information.

Delineation of groundwater potential zones in semi-arid region of Jiledubanderu river basin-Hydroresearch, Science Direct journal, Anantapur District, Andhra Pradesh, India using fuzzy logic, AHP and integrated fuzzy-AHP approaches, 26 November 2019.

Water is a renewable and essential source of human feasibility and various ecosystems, so this study focused on the delineation of groundwater potential zones in the semi-arid region of Jiledubanderu river basin, Anantapur District Andhra Pradesh, India using fuzzy logic AHP and integrated fuzzy-AHP approaches.

This paper study identifies GWPZ for groundwater existences by analyzing different thematic layers as influencing factors. In this study a three-way validation is performed to verify the accuracy of the GWPZ maps.

Hydrogeological delineation of groundwater vulnerability to droughts in semi-arid areas of western Ahmednagar district- Egyptian journal of remote sensing and space sciences, Elsevier journal, 7December 2016.

The paper is based on the hydrogeological delineation of groundwater vulnerability to droughts in semi- arid areas of the western Ahmednagar District.

This paper gives a brief idea about the groundwater scarcity zones providing detailed methodology to prioritize areas evitable to groundwater unavailability based on different influencing thematic layers, by characterizing the study area into various evitable class types of extreme, high, moderate and low.

AkvaGIS: an open-source tool for water quantity and management. Computers and geosciences ELSEVIER journal, Institute of environmental assessment and water research, Spain, 2019.

The geographical information system provides useful tools for addressing the physical and chemical parameters in collection analysis and visualization data in different formats. This software is widely used by the scientific community and the private sector. Also, this platform may aid in reducing environmental assessment such as the evolution of water quality water availability zone mapping from the local to regional scale. In this paper we have presented AkvaGIS open-source GIS based package integrated into the FREEWAT platform. This software manager and visualizes hydrogeological and hydrochemical data and prepare to hydrogeological input files for any groundwater in all available formats in QGIS. Also, AkvaGIS can be updated by any advanced user.

Watershed delineation using GIS for supporting the integrated and sustainable watershed management case study: Research Gate journal, Manjuntio Watershed, Bengkulu Province, Indonesia, 27 July 2016.

Improper management of land use activities has led to losses in economic and environmental and reducing the impacts of integrated and sustainable water management in Indonesia is urgently required.

Lack of accurate data, especially outside Java, has made it difficult to synchronize, integrate and co-ordinate the watersheds. Integrated and sustainable watershed management requires efficient and effective implementation agencies at the utilization of Remote sensing and GIS is considered to be the most favorable option. The study is to determine the boundary areas and flow pattern of the watershed named Manjuntio province in Bengkulu using GIS.

The study will be used for progressing research namely the development of integrated and sustainable water management which considers both physical and socio-economic aspects.

Morphometric Analysis of Baltira Watershed using QGIS platform- International journal of recent technology and engineering, volume 8 Issue-2, July 2019.

In this paper, the author analyzed the Baltira watershed using QGIS platform by morphometric analysis. The study says that QGIS is used for detailed morphometric assessment for the Baltira watershed covering an area of 273 sq.km area of the tributaries of man river. The watershed of Baltira has reasonably homogeneous geology having high relief and steep ground slope. By the calculation of hydrological module QGIS software, watersheddelineation and drainage parameter are studied by utilizing ASTER- DEM.

Watershed Delineation of Khumbhoj Nala Basin by using QGIS- International research journal of engineering and research, Volume 6 Issue-4, April 2019.

This paper analyses Kumbhoj Nala watershed is delineated by using QGIS software.

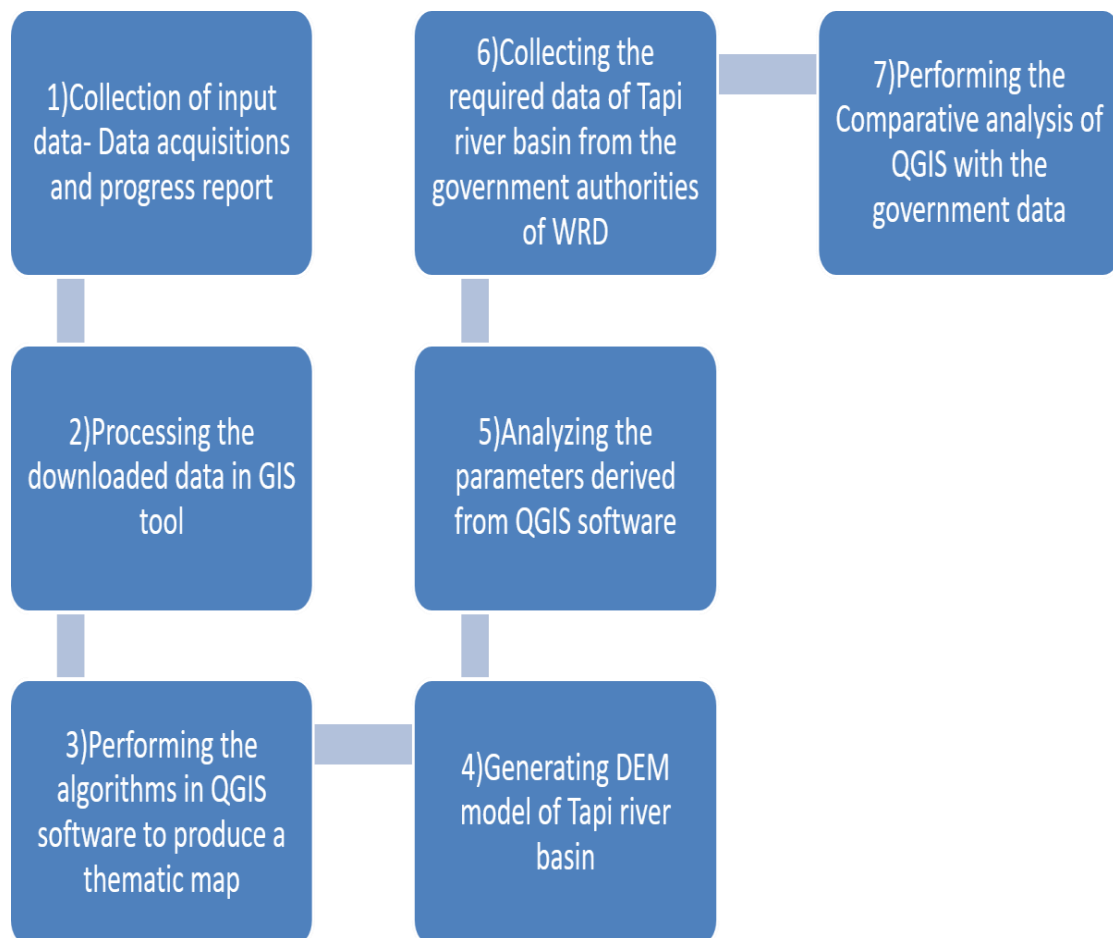
With a wide range of applications, the GIS tool comes with grasp interphase and user-friendly commands. Various studies on soil resource management and watershed have been performedusing the GIS modeling. It is a computationally efficient tool and is computer-based modelling software. It includes various inputs like vectors and raster. It allows studying long- term impacts. As the water table goes on decreasing day by day watershed delineation becomes vital for nalas of various villages.

CHAPTER 3

METHODOLOGY

3.0 Introduction: This section gives a brief explanation of the defined methodology. The defined methodology includes total work planned as per month-wise and the work completed till now.

3.1 Defined methodology-total work planned:



3.2 Month wise planning of work Completed:

Table 3.1: Month wise planning of work completed.

Month	Project Work
August	• To determine the study area of the project • Collection and to import the toposheets and shapefiles.
September	• Collection of input data- Data acquisitions and progress report
October	• Progressing the downloaded data in GIS tool
November	• Performing the algorithms in QGIS software to produce a thematic map
December	• To generate a map of watershed delineation of Tapi river basin using QGIS
January	• Generating DEM model of Tapi river basin
February	• Collecting the required data of Tapi river basin from the government authorities of WRD
March	• Performing the Comparative analysis of QGIS with government data • Preparation of final report

3.3 QGIS Stepwise Methodology:

3.3.1 Collection of Input data

University of Texas library provides entire topographical sheets for India and also neighboring countries. Image format – JPG format Scale - 1:2,50,000

We have collected the toposheets of all the Tapi river cities such as Surat, Burahanpur, Bhusawal, Betul, Songadh, Dhule, Nashik, Amravati, Jalgaon, Khandwa, Malegaon, Buldana, Washim, Nandurbar

Portal link–<https://legacy.lib.utexas.edu/maps/arms/india/>

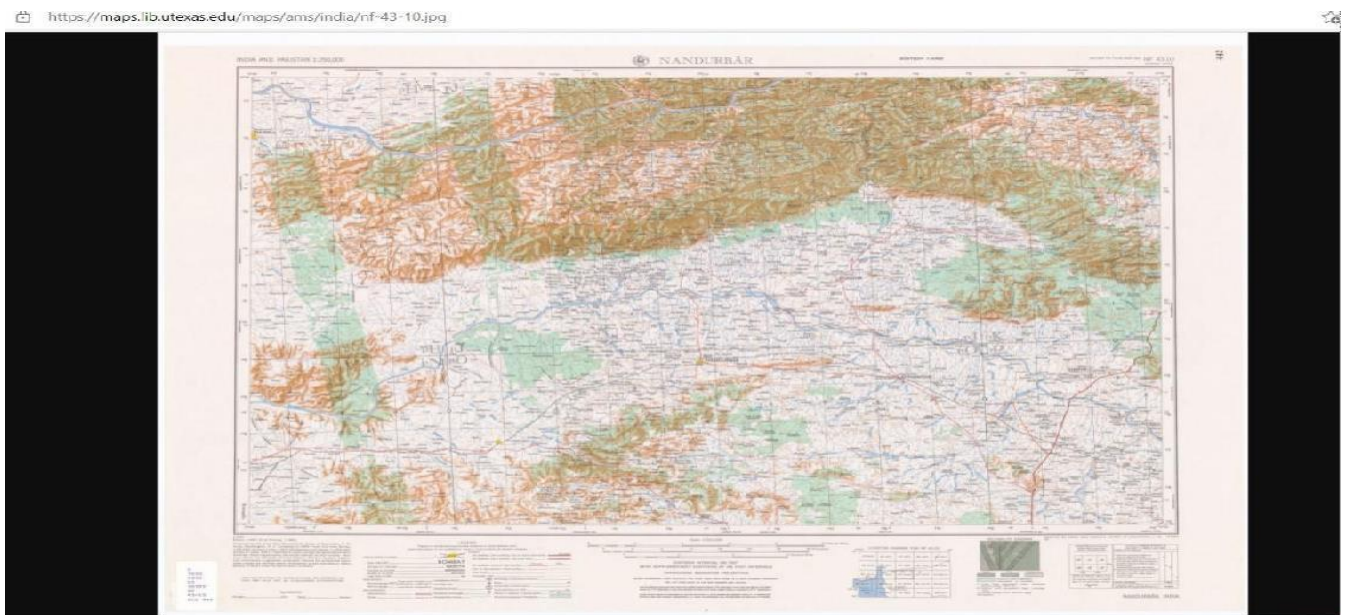


Fig. 3.1: Toposheet of Nandurbar

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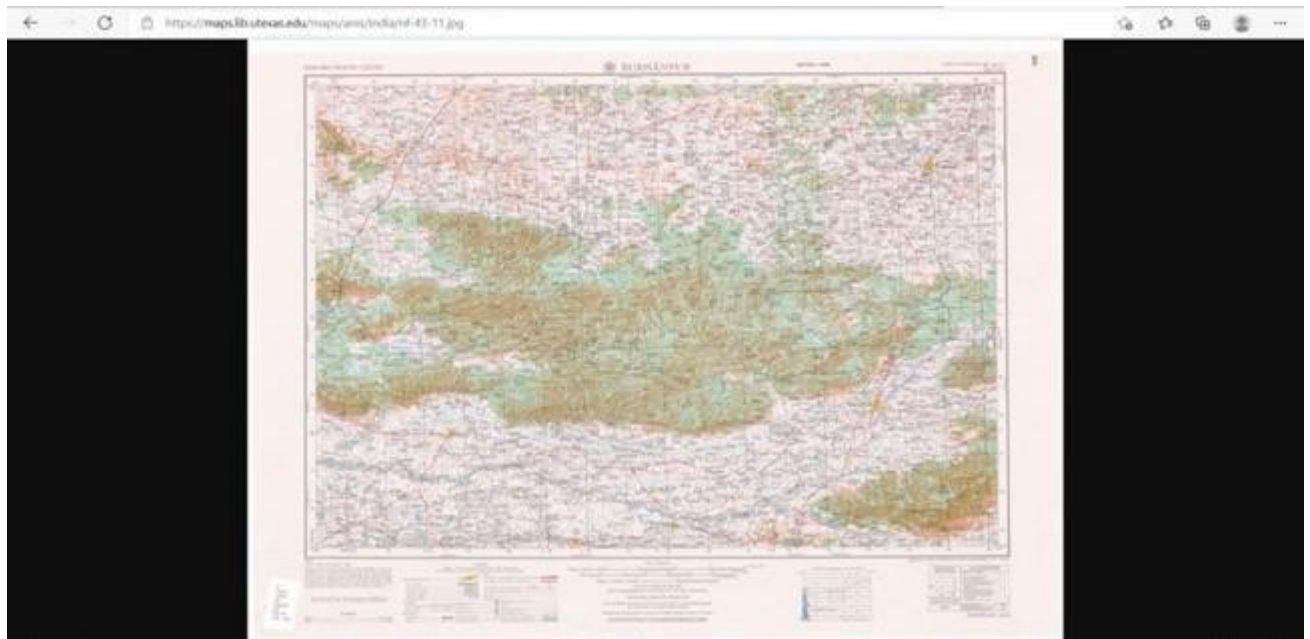


Fig. 3.2: Toposheet of Burhanpur

- Collection of satellite image – Bhuvan Portal**

ISRO has developed a web base mapping application called Bhuvan portal. It provides: several spatial data for different time periods.

Portal Link – <https://bhuvan.nrsc.gov.in/bhuvan>

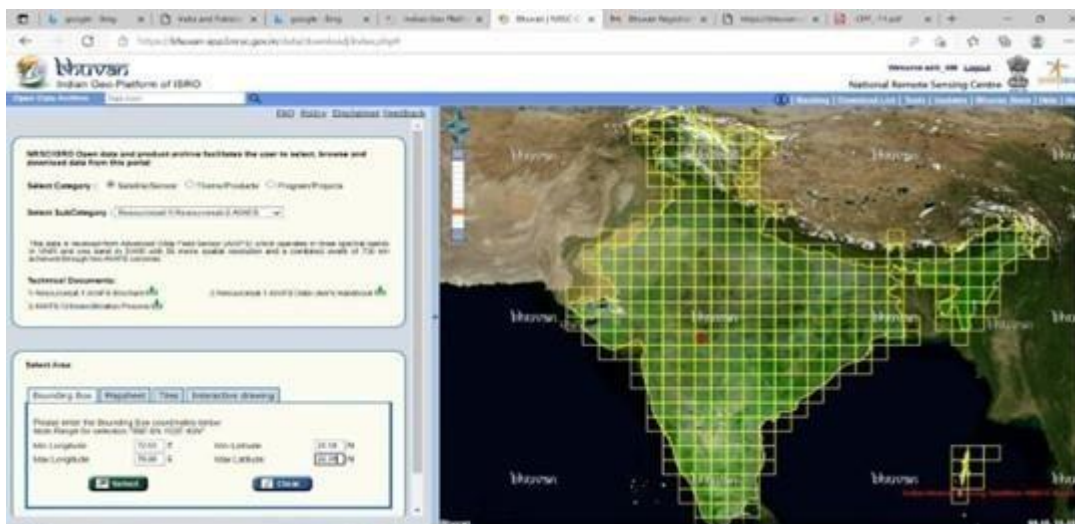


Fig. 3.3: Satellite image from bhuvan portal

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- **Collection of satellite image – Earth Explorer Portal**

We can download sequential data for temporary analysis.

It provides a long list of satellites and aerial images.

Portal link: <https://earthexplorer.usgs.gov/>

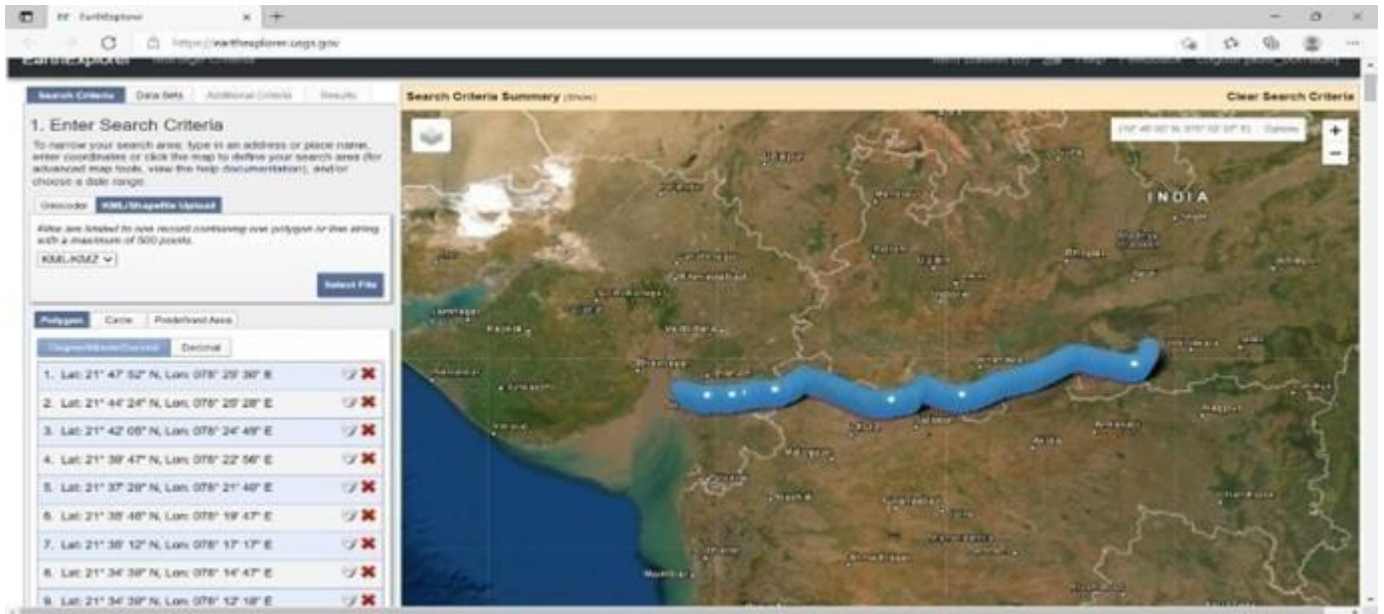


Fig. 3.4: Satellite images of Tapi river catchment area

- **SRTM data Collection – Opendem**

Earth surface elevation information was collected by several radar topographical missions in year 2000. Data was collected from the region of 56 S latitude to 60 N latitude and now this information is given in the form of contours.

- **Vector Data Collection from Diva GIS**

This portal provides open-source software and spatial data for the entire world. This is a popular tool for downloading vector data.

3.3.2 Processing the downloaded files in QGIS tool

- Search the area of interest or upload the KML shapefile to generate a location map and SRTM DEM file

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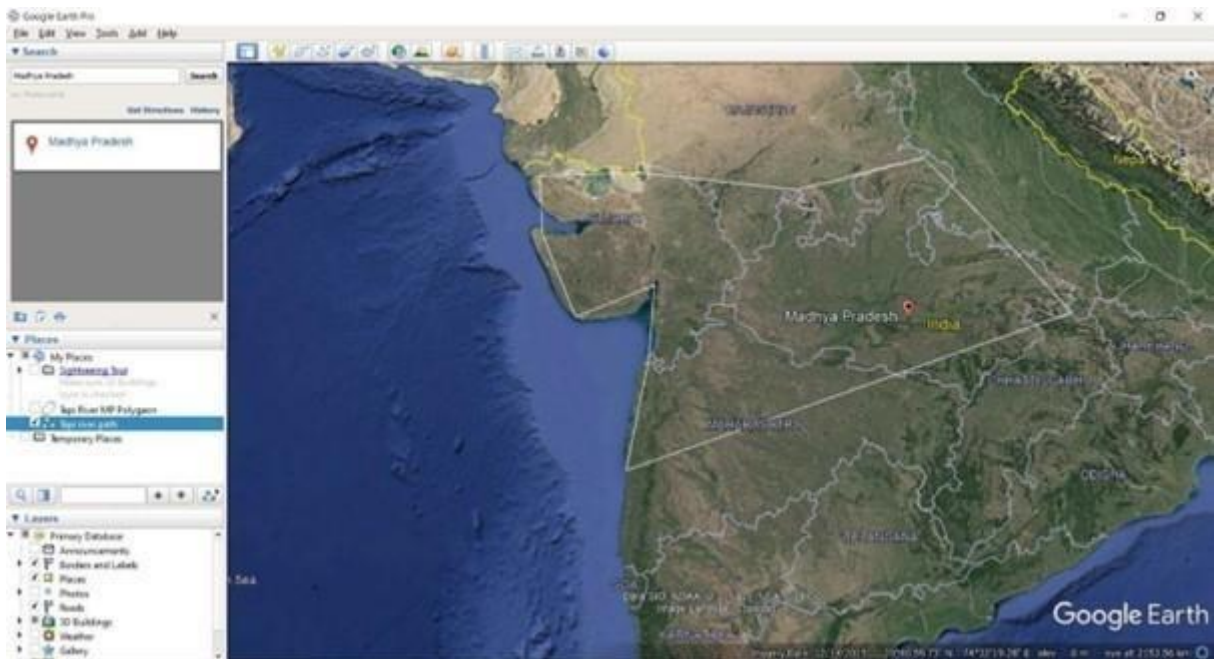


Fig. 3.5: Earth explorer file showing the study area

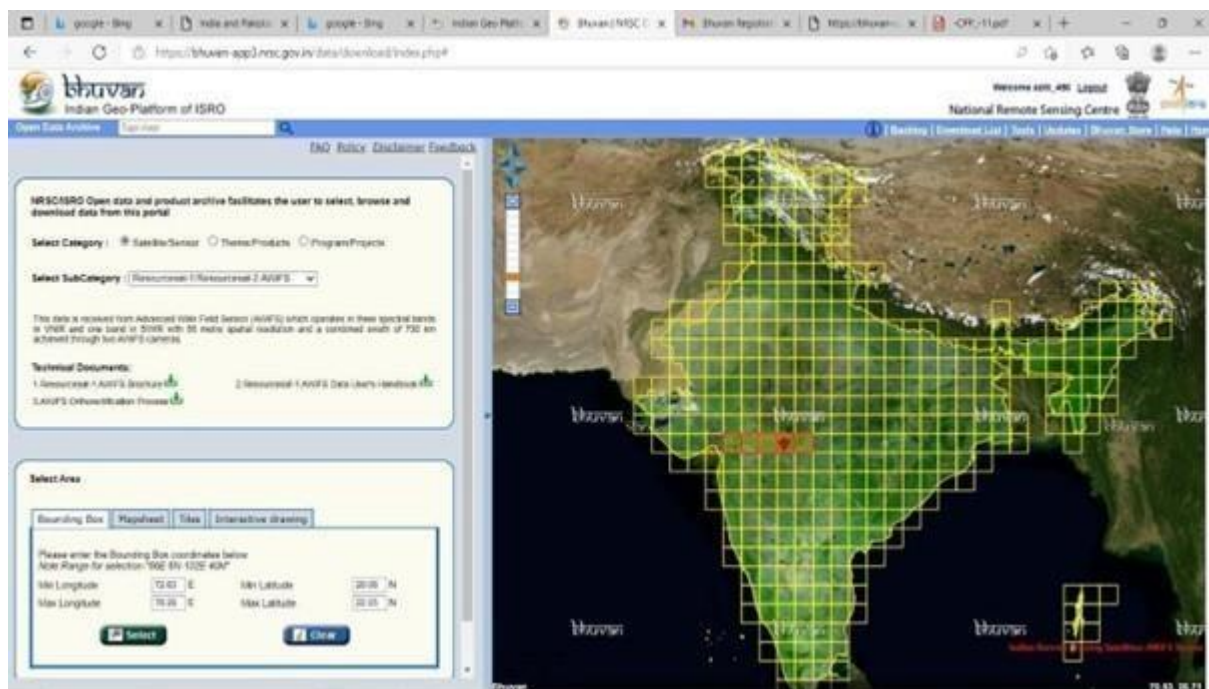


Fig. 3.6: Bhuvan portal location map of Tapi river catchment area

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- Install the coordinator plugin, HCMGIS plugin and add raster layer to generate the reprojected DEM file.

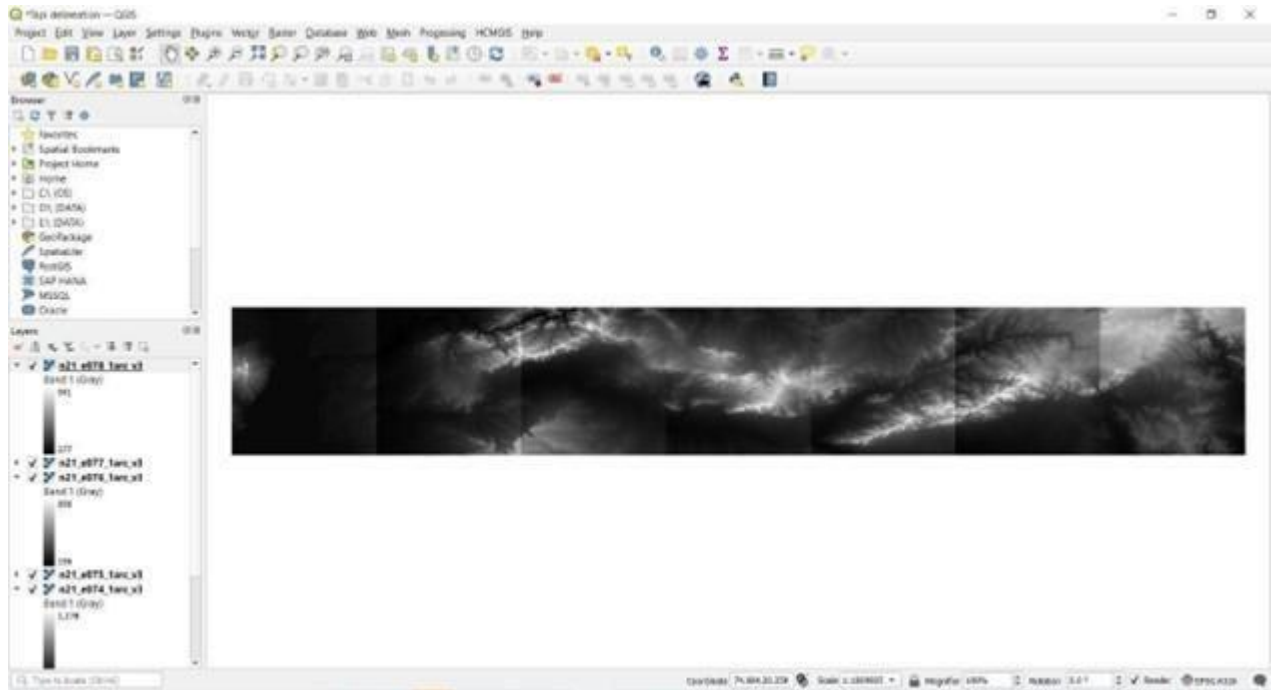
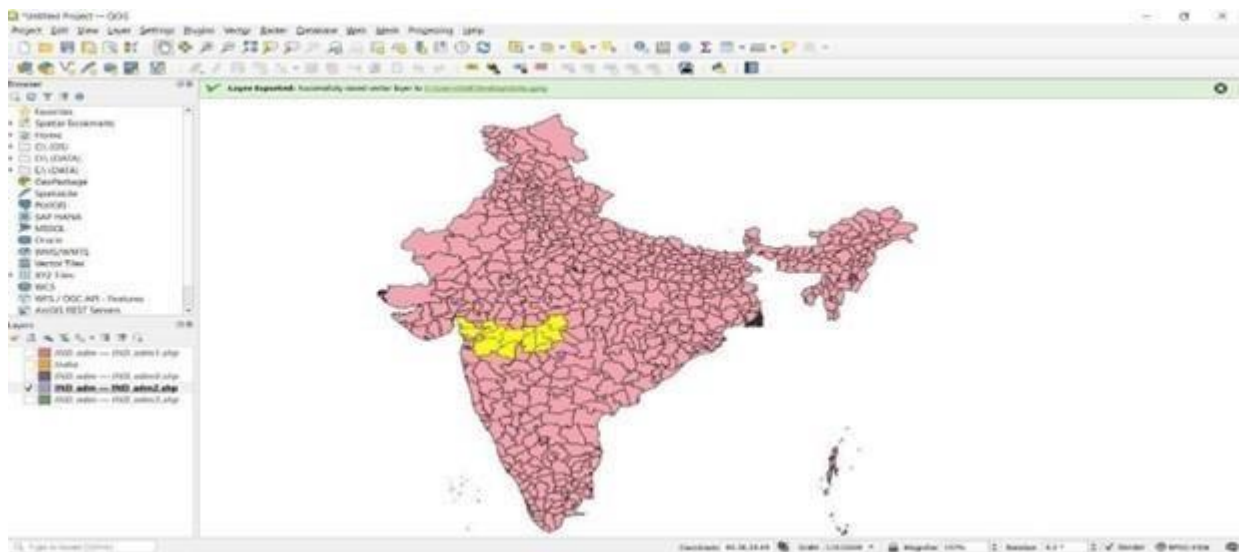


Fig. 3.7: QGIS Reprojected DEM file



- Processing the vector data collected from DIVA GIS and adding vector layer to generate a location map of Tapi river basin.

Fig. 3.8: Location map generated in QGIS

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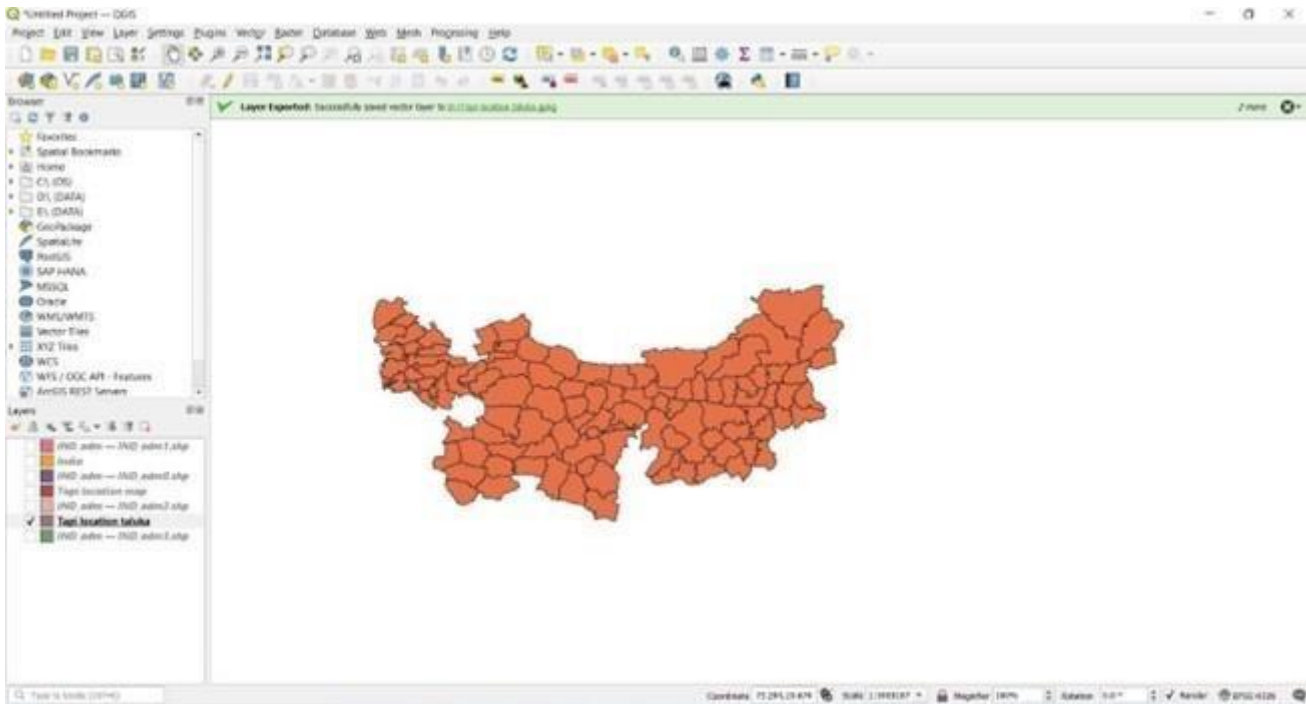


Fig. 3.9: Location map showing detailed Tapi river watershed area

3.3.3 Performing the algorithms in QGIS software to produce a thematic map.

- Fill the sinks (wand and liu) from SAGA toolbox to generate Strahler stream order.

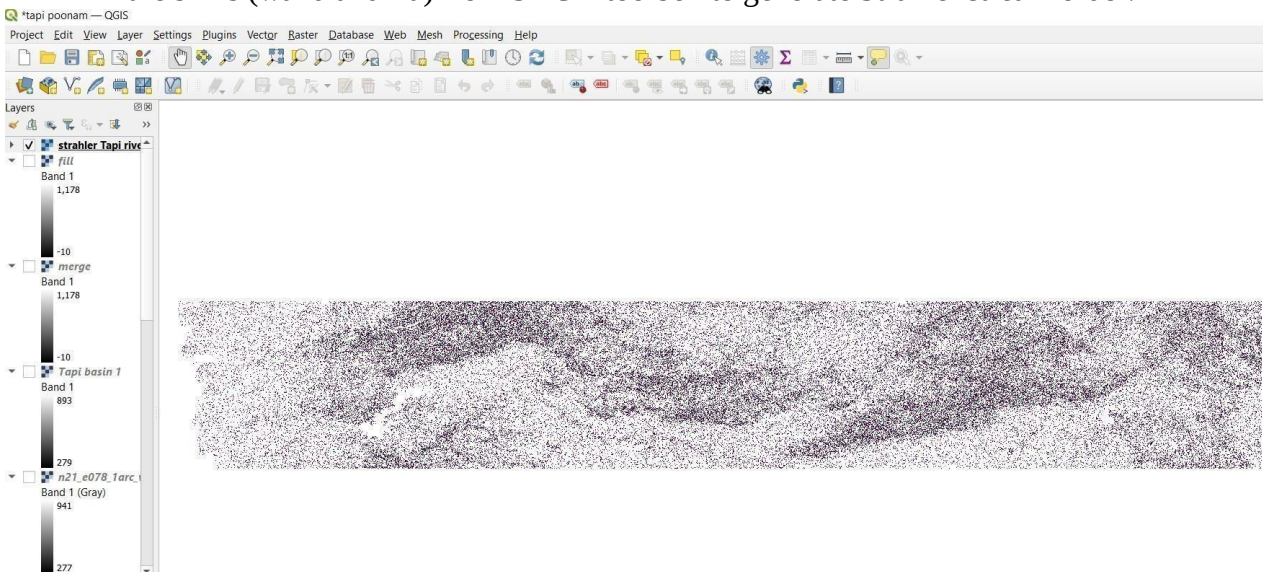


Fig. 3.10: Viridis colour ramp showing Strahler stream order.

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- Identification and addition of river channel order layer and determine the flow direction

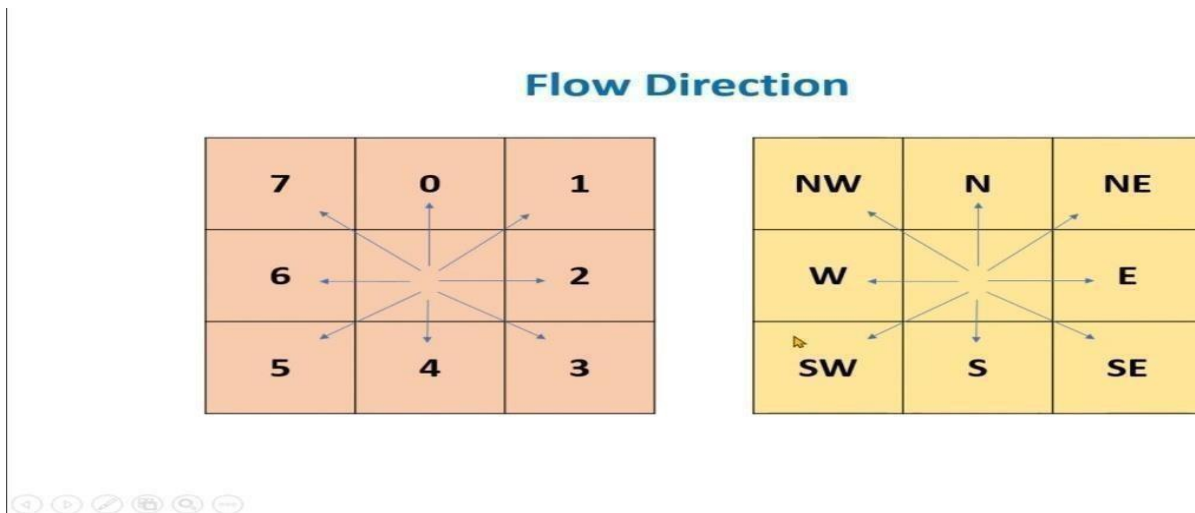


Fig. 3.11: Flow direction with respect to stream order

- Create the outlet points and generate the outlet points.

3.3.4 Generating final thematic model of watershed delineation

- Arrange the layers as – flow direction, channels, outlet points, upslope area, drainage basin, Strahler stream order, google satellite, filled DEM, reprojected DEM, shape files.
- Process the raster with polygon function in the processing toolbox.

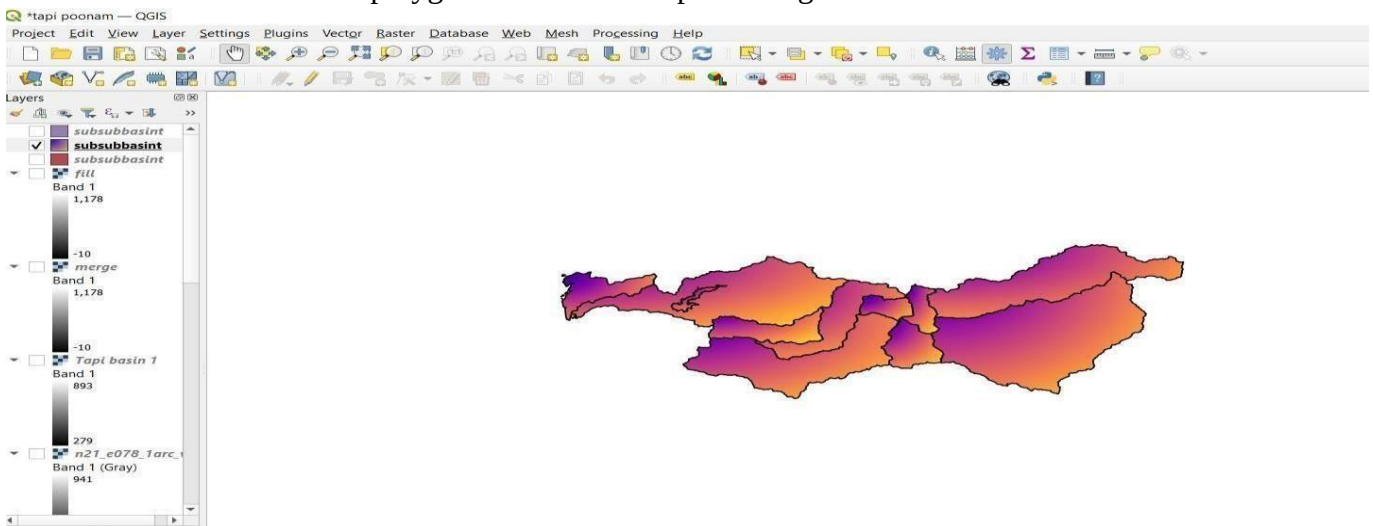


Fig. 3.12: QGIS DEM model of Tapi river basin

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Discharges at GD sites in the Tapi basin:

The catchment area of Tapi Basin in Maharashtra is 55101.54 sqkm. There are 24 Gauge stations in this catchment. The following table shows discharges for some GD stations in this catchment.

Table 4.1: Discharge at GD sites in Tapi river basin

Sr.No	GD site	June	July	August	September	October	Total Discharge In Cumec
1	Aurangpur	-	15.23	21.65	15.28	3.45	55.62
2	Bhusawal	-	413.41	765.48	803.10	13.05	1895.04
3	Daryapur	-	-	1.01	0.79	0.61	2.41
4	Fardapur	1.72	-	-	2.83	3.52	8.06
5	Jamner	3.65	-	13.27	27.25	17.43	61.59
6	Khariya	0.35	298.75	340.88	461.73	15.84	1117.56
7	Malegaon Girna	-	59.12	49.31	50.84	31.74	191.02
8	Manasgaon	47.96	77.18	138.44	168.83	139.05	571.45
9	PalTapi	-	-	69.66	74.14	-	143.80
10	Sukwad	-	192.14	200.26	325.97	129.61	847.98
11	Suple	-	79.99	119.14	54.69	32.82	266.65
12	Tonda	-	12.70	25.16	11.68	15.61	65.16

***“DELINEATION OF WATERSHED AND ESTIMATION OF DISCHARGE OF
RIVER TAPI SUB-BASIN BY USING GIS TOOLS”***

4.2 Discharges by QGIS at GD sites in the Tapi basin:

Table 4.2: Discharge by QGIS at GD sites in Tapi river basin

Sr.No	GD site	June	July	August	September	October	Total Discharge In Cumec
1	Aurangpur	-	12.94	18.40	12.98	2.93	47.27
2	Bhusawal	-	351.39	650.65	682.63	11.09	1610.78
3	Daryapur	-	-	0.858	0.67	0.51	2.04
4	Fardapur	1.46	-	-	2.40	2.99	6.85
5	Jamner	3.10	-	11.27	23.16	14.81	52.35
6	Khariya	0.29	253.93	289.74	392.4	13.46	949.9
7	Malegaon Girna	-	50.25	41.91	43.21	26.98	162.32
8	Manasgaon	40.77	65.60	117.67	143.51	118.19	485.74
9	PalTapi	-	-	59.21	63.01	-	122.23
10	Sukwad	-	163.32	170.22	277.07	110.17	780.18
11	Suple	-	67.99	101.27	46.29	27.90	243.65
12	Tonda	-	10.80	21.39	9.93	13.27	55.39

***“DELINEATION OF WATERSHED AND ESTIMATION OF DISCHARGE OF
RIVER TAPI BY USING GIS TOOLS”***

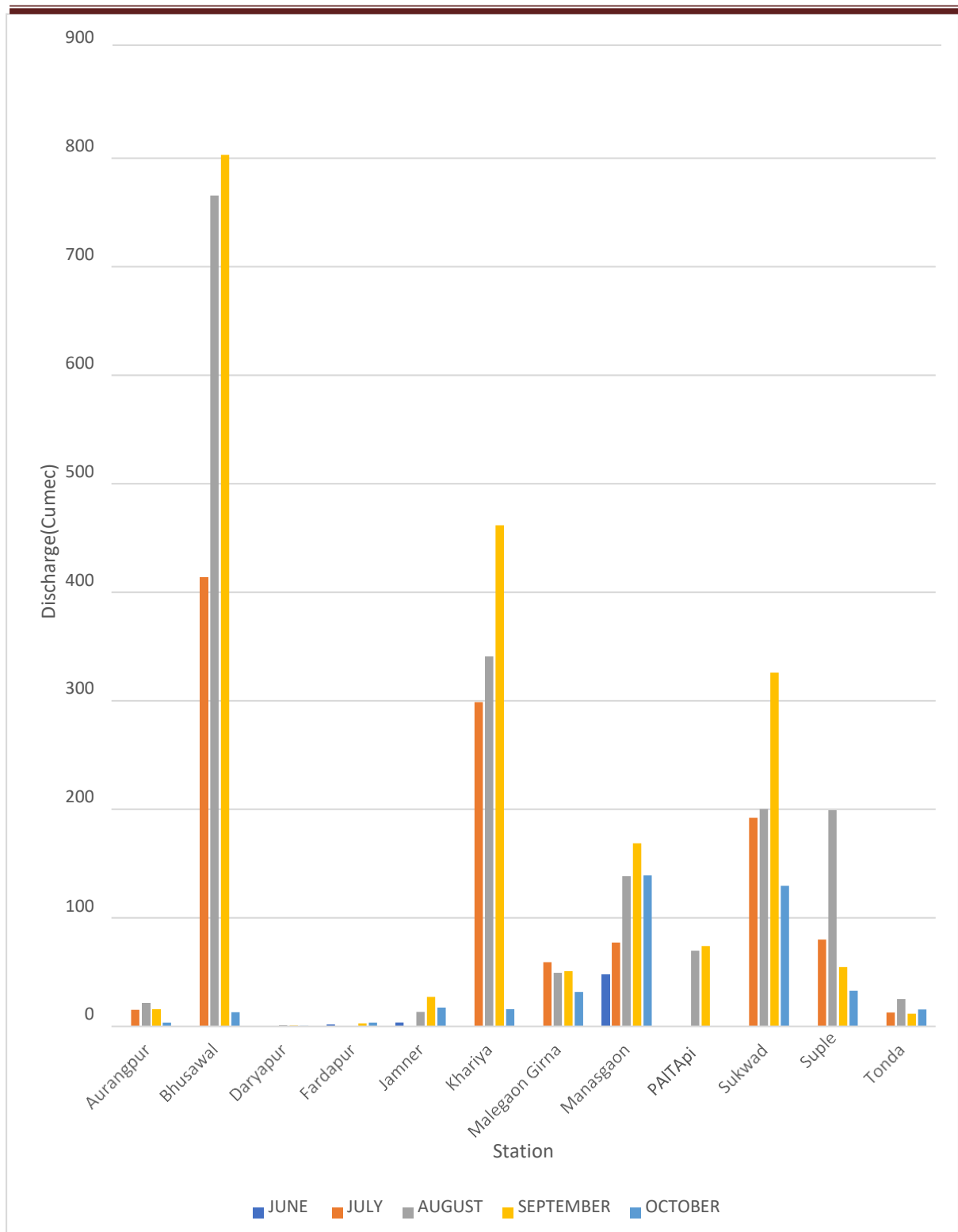


Fig. 4.1: Discharge at GD sites in Tapi river basin

***“DELINEATION OF WATERSHED AND ESTIMATION OF DISCHARGE OF
RIVER TAPI BY USING GIS TOOLS”***

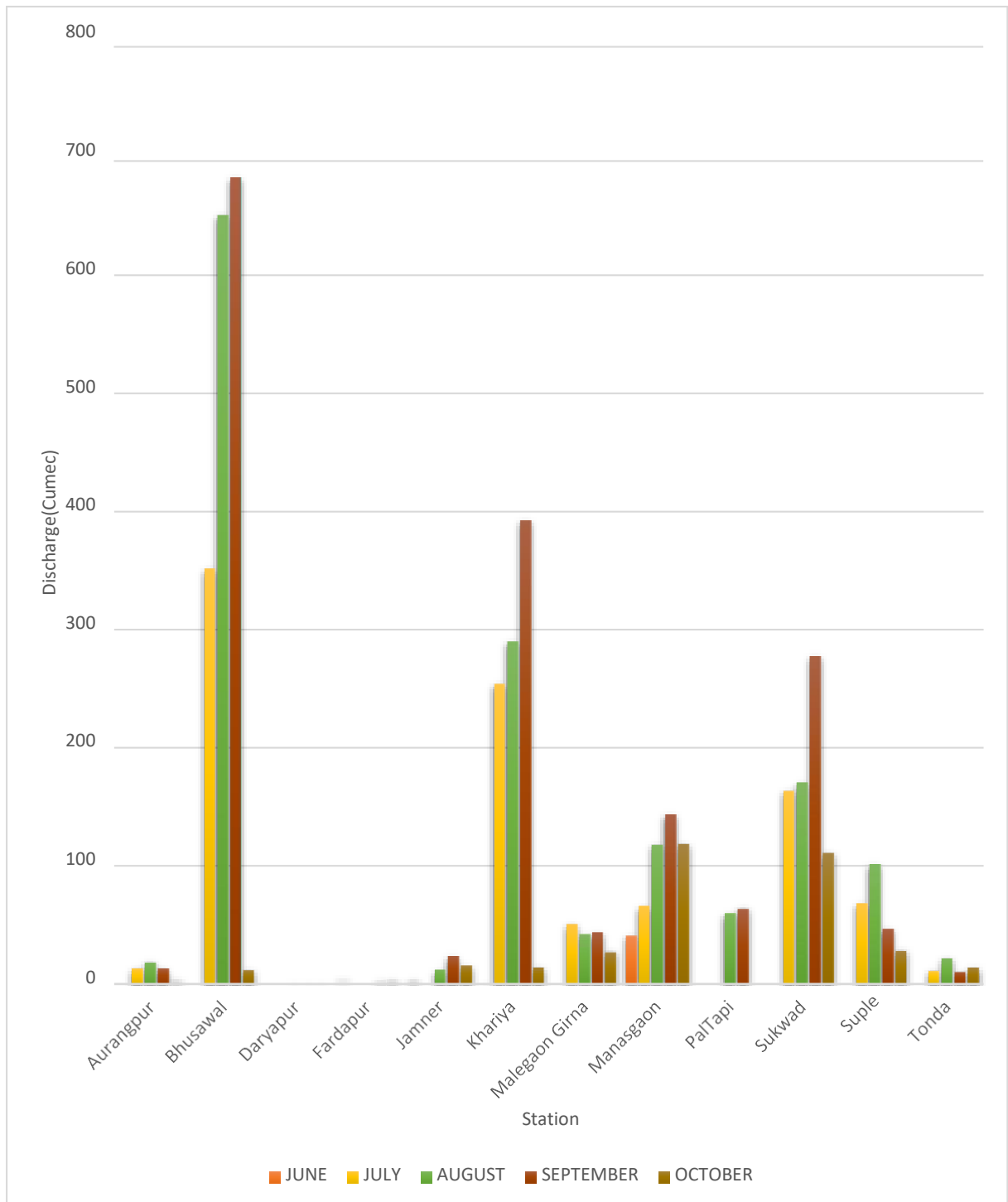


Fig. 4.2: Discharge by QGIS at GD sites in Tapi river basin

4.3 Isohyets of Tapi river basin

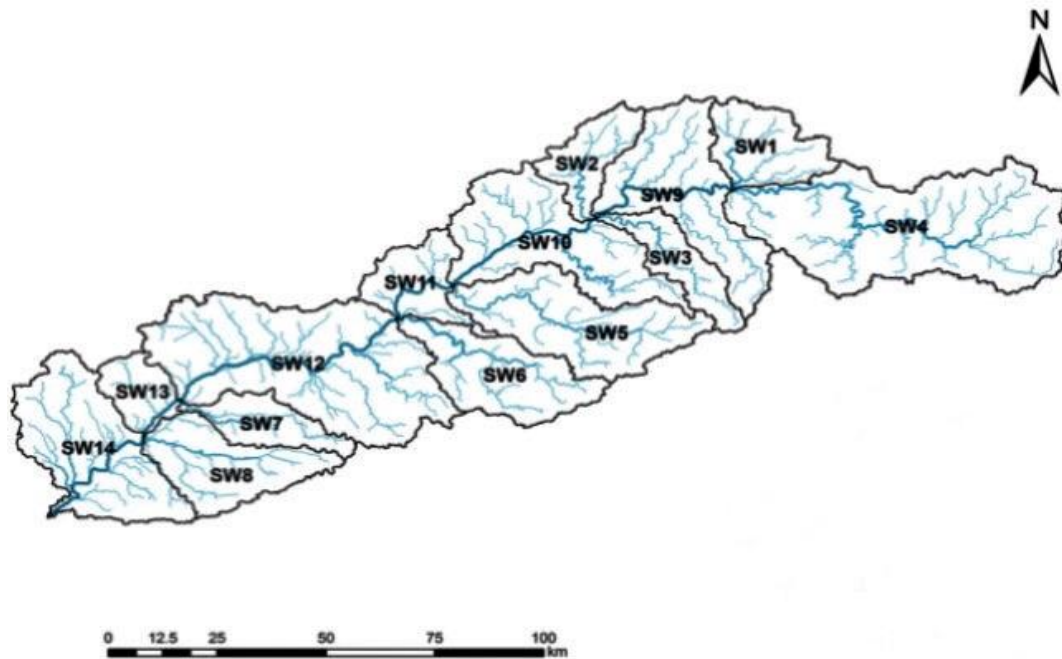


Fig. 4.3: Stream network of Tapi river basin

4.4 Shapefiles generated in GIS software:

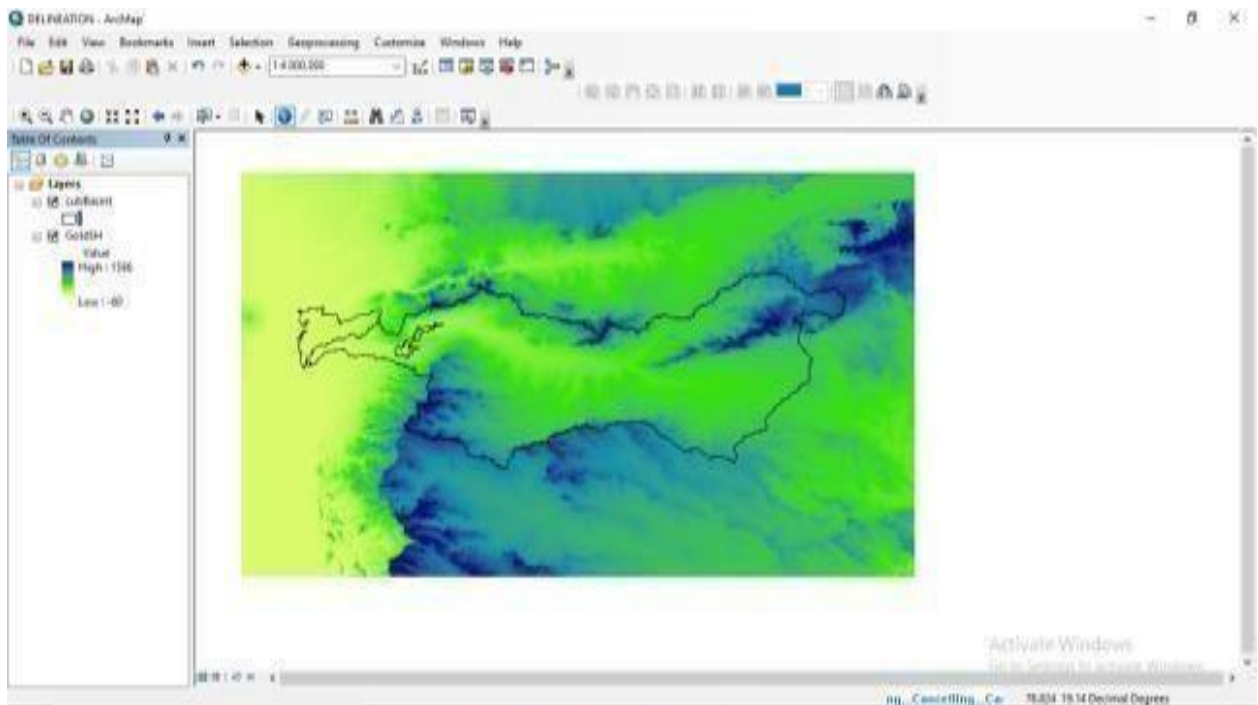


Fig. 4.4: Elevation shapefile of Tapi river basin

“DELINEATION OF WATERSHED AND ESTIMATION OF DISCHARGE OF RIVER TAPI SUB-BASIN BY USING GIS TOOLS”

4.5 Digital Elevation Model of Tapi Basin:

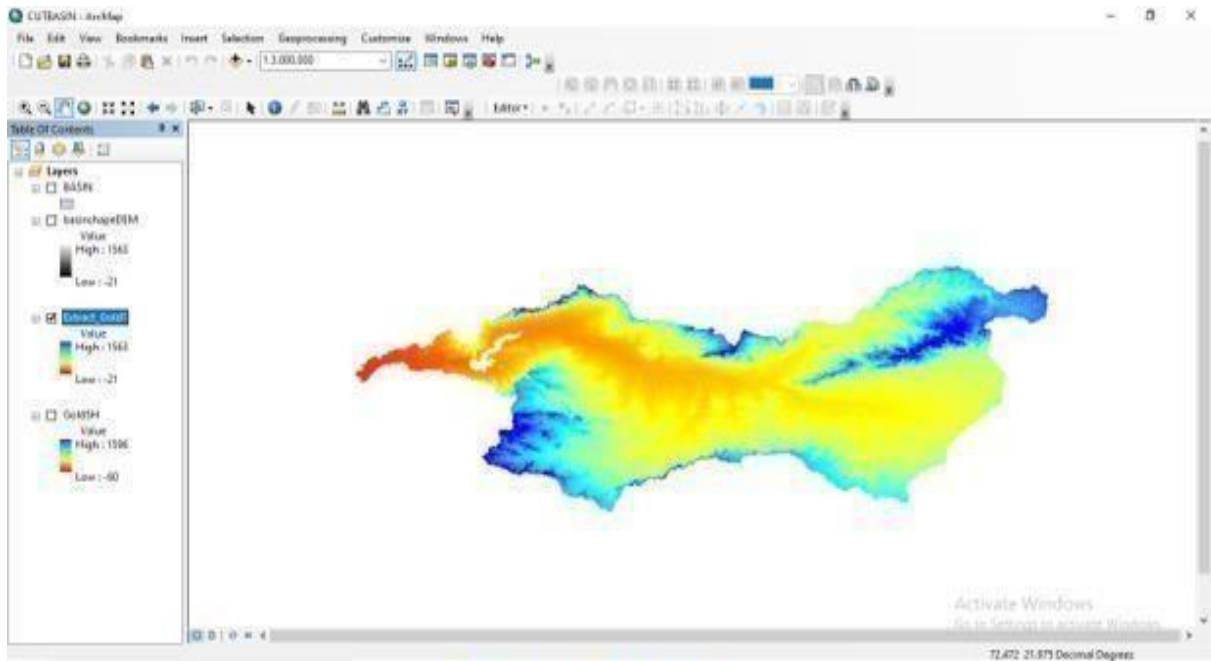


Fig. 4.5: 2D Digital Elevation Model of Tapi Basin

4.6 3D Elevation Model of Tapi Basin:

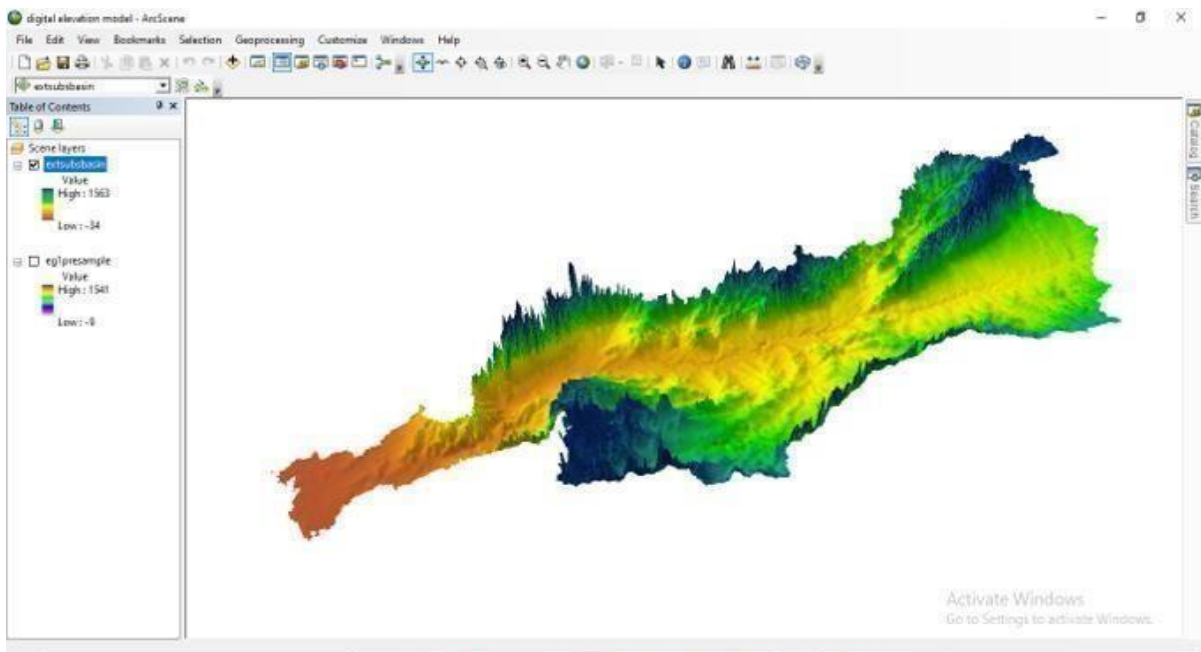


Fig. 4.6: 3D Digital Elevation Model of Tapi Basin

APPLICATIONS TO THE REAL WORLD

1. Waste management:

Waste management includes processes and actions required to manage waste from its inception through its final disposal.

This includes the collection, transport treatment and disposal of waste together. with monitoring and regulation of waste Management process and waste-related laws technologies economic mechanisms.

Waste management practices are not uniform among countries and like residential and industrial sectors and all take different approaches.

Proper management of waste can be done by using GIS which is important for building sustainable and life-able cities.

2. Crime mapping:

Crime mapping is used by analysts in law enforcement agencies to map, visualize and analyze crime incident patterns.

It is a key component of crime analysis and economics and policy strategy mapping crime using geographic information system (GIS) allows crime analysis to identify crime hot spots along with other trends and patterns.

3. Urban planning:

Urban planning is a technical and political process that is focused on the development and design of land use and built environment including air water and the infrastructure passing into and out of urban areas such as transportation, communication and distribution networks and their accessibility.

Traditionally, Urban planning followed a top-down approach in master planning the physical layout of human settlements. The primary concern was the public welfare, which included considerations of efficiency, sanitation, protection of the environment as well as effects of the master plan on the social and economic activities.

Over time, by using GIS urban planning has made it easy for locating and improving road development throughout the country.

4. Flood maps:

A flood map informs your community about the local flood risk. It helps set minimum floodplain standards so that your community builds safety and resilience. It determines the cost of flood insurance which helps property owners to financially protect themselves against flooding. The lower your degree of risk, the lower your flood insurance premium will be. In areas with a high risk of flooding, you might be required to get flood insurance. To ensure the public knows it is necessary to predict it before taking place and this could be done effectively using GIS through this collaborative process a community can give you appeal and contribute to the development of flood maps before it is adopted by the community.

5. Remote sensing applications:

Remote Sensing software is a software application that processes remote sensing data. Remote Sensing applications are similar to graphics software, but they enable generating geographic information from satellite and airborne sensor data. Remote Sensing applications read specialized file formats that contain sensor image data, georeferencing information, and sensor metadata.

6. Traffic data and Census data:

The GIS software will be analyzing the spatial data of different types of traffic accidents and census data for a given set of geographic areas (such as local authorities, wards or super output areas), boundary data for the same set of areas etc. It is carried out on an annual basis because that is when year-end traffic collision and volume data is available. The files used in this tutorial are polygon shape file. You may use any other data for another city or location as long as there is a polygon shape file of your chosen city or neighborhood, a polyline shape file for roads, and a point shape file for accidents, as well as any other information you would like to analyze.

7. Environmental Information:

This software will assist in receiving the mandatory skills in the manipulation of QGIS tools that are suggested in environmental studies and analysis (EIA, environmental papers, fauna studies, etc.). It will be guided in the development of standard thematic cartography, and multi-criteria spatial analysis with relevance in environmental management and in substitute analysis. Also, to mark the significance and advantage of a Geographic Information System (GIS), its integration and applicability in environmental management. Execution of environmental GIS projects, and their solutions through practical exercises. Management of field sampling spatial databases, fauna density maps, erosion studies, hydrological studies, and environmental sensitivity maps.

8. Electricity Network Management:

QGIS software has been entitling the electric utility industry for decades. It gives a robust framework for asset and workforce management in addition to tools for drafting and study. It's influential to have a correct asset inventory with electric poles, transformers, cable lines, substations and other electric utility assets properly mapped. GIS permits us to arrange the asset data a lot easier and helps us to extract insight from the infrastructure.

With a properly setup of software, you can get good insights from infrastructure data by studying usage patterns, realizing problems and risks like power outages, having oversight about energy consumption, searching for potential threats to the power grid, and managing utility asset repairs.

9. Cadastre Management:

Cadastre managers and GIS professionals can use the GIS parcel data model as a initial point for defining parcel information and for executing migration strategies from existing data designs to a geodatabase environment. The data model gives a quicker, economical solution to data migrations, the extensive and expensive part of a GIS project. Decision makers can apply the model to integrate landownership information with other data. Land and GIS professionals can try the definitions and structure of the model to establish consistent and representative parcel information for data distribution. The desire of the GIS parcel model is to narrate parcel information to help local government and private sector decision making. ESRI's GIS software helps and is agreeable with the Cadastre Data Model by the International Federation of Surveyors.

CHAPTER 5
CONCLUSION

5.1 Conclusions

- This study gives a brief idea of watershed delineation using the GIS tool. The watershed is delineated by performing algorithms in QGIS software to produce a thematic map of a catchment area 65,145km².
- The morphometric parameters which are derived from the DEM model are one of the most important sources of spatial information in the raster form. This raster information is basically in the matrix of square areas to define the location of features in the form of pixels/cells.
- **Loading and error issues:**

QGIS	
1. QGIS needs less time to load and process files.	
2. We have faced so many errors during delineation work. E.g. Invalid geometry files, invalid vector files.	

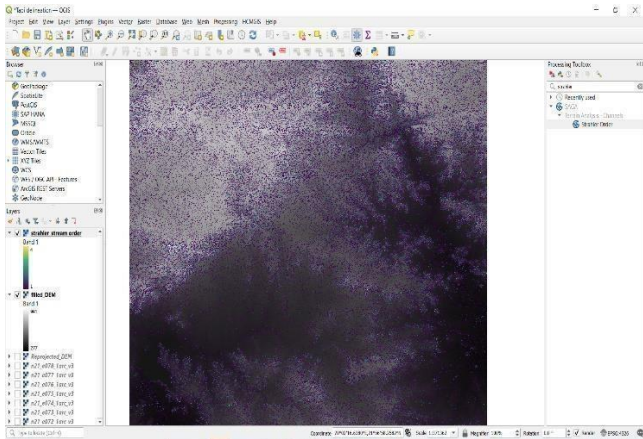
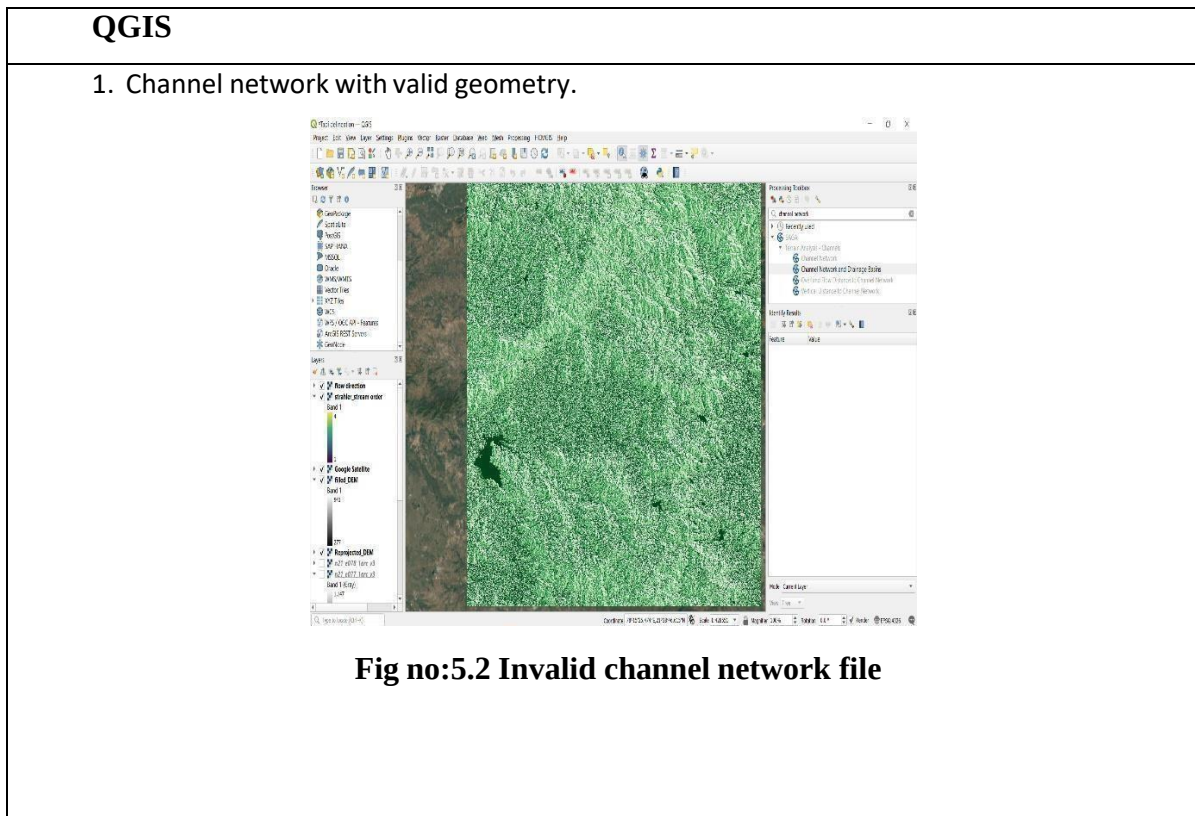


Fig no: 5.1 Invalid vector file

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- **Invalid geometry issues:**



- We didn't get the 3D model in QGIS.

5.2 Future Scope

- GIS software is easy to use and gives more satisfactory results in a short time as compared to manual planning,
- The study of watershed delineation by using GIS tool will be the best reference for the following parameters:
 1. Selecting sites for water harvesting structures.
 2. Adoption of water conservation measures.
 3. Making policies by local government authorities related to water resources, irrigation and disaster management system.
 4. A detailed estimation of cost for rejuvenation of system of various orders.

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